

SOUBHAGYA SADHUKHAN

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EDUCATION

Institute of Engineering & Management, Kolkata

2023 – 2027

B.Tech in Computer Science & Engineering (AI and Machine Learning)

TECHNICAL SKILLS

Languages: Python, C, Java, Solidity, JavaScript, SQL

Frameworks & Tools: React, Next.js, Node.js, FastAPI, Flask, Docker, Git, AWS (EC2, S3)

ML & Web3: Gemini API, RAG Pipelines, Prompt Engineering, zk-SNARKs, Algorand, Circom

INTERNSHIPS

AgentMesh agent-mesh.app

June 2026 – Present

Co-Founder

- Built a no-code visual canvas for composing autonomous AI agent pipelines; each agent node holds a real Algorand wallet and pays for API calls via the x402 micropayment protocol.
- Architected Next.js + FastAPI stack with SSE streaming, AES-GCM encrypted mnemonics, OAuth auth, and five LLM providers; 2% take-rate on x402 calls as core revenue model.
- Won **Algorand Track** at AlgoBharat HackSeries 3.0 Dev Retreat (Goa, June 2025); **Avalanche Foundation** startup grant recipient.

Avacado avacado.app

October 2025 – Present

Founding Engineer

- Built a privacy-focused DeFi platform using zk-SNARKs (Groth16), ElGamal encryption, and Poseidon hashing for shielded on-chain balances.
- Designed Solidity + Circom zero-knowledge circuits, optimizing proof generation time and on-chain verification gas costs.

PROJECTS

AlgoGate | *Python, FastAPI, Algorand, PyPI, WebSocket* | [Repository](#)

- Published **allogate-sdk on PyPI** – one-decorator API route protection via Algorand x402 micropayments, enabling pay-per-use monetization without API key infrastructure.
- Built Algorand indexer-based payment verification with JWT session tokens, nonce tracking, and replay-attack protection; shipped live WebSocket dashboard for real-time monitoring.
- Won **Algorand Track** at BinaryV2, KGEK.

llm-vcs | *React, Node.js, LLM APIs* | [Repository](#)

- Built a GPT-style chat platform with **Git-like version control** for prompt engineering – commits, branching, rollback, and full history traversal across prompt sessions.
- Designed a reproducible prompt pipeline enabling structured LLM experimentation with diff and comparison across versions, addressing core reproducibility gaps in AI development.

Entropy TicTacToe | *React, TypeScript, FastAPI, WebSocket* | [Repository](#)

- Designed a real-time multiplayer strategy game with hidden-information mechanics – players don't know which symbol they control until reveal.
- Implemented dynamic probability updates using the **Monty Hall theorem**, adding an information-theoretic decision layer with posterior recalculation on each reveal.
- Architected server-authoritative game logic over WebSockets (FastAPI) with two-phase gameplay, room-based matchmaking, and a React/TypeScript frontend on Vercel.

ACCOMPLISHMENTS

- **ETHGlobal New Delhi 2025** – 1inch Track Winner
- **AlgoBharat HackSeries 3.0 Dev Retreat** – 2nd Place, Algorand Track (AgentMesh)
- **Avalanche Foundation** – Startup Grant Recipient